House Price Prediction — Summary of Findings

Approach

The dataset was cleaned (no missing values or duplicates were found), categorical yes/no fields were converted to binary, and the furnishing status field was one-hot encoded. The data was split 80/20 into training and test sets, and two regression models — Linear Regression and Random Forest — were trained to predict price from the remaining 12 features.

Model Performance

Model	MAE	RMSE	R^2 Score
Linear Regression	₹9,70,043	₹13,24,507	0.653
Random Forest	₹10,14,947	₹13,99,769	0.612

Linear Regression performed slightly better than Random Forest on this dataset, suggesting the relationship between features (especially area) and price is largely linear. Both models explain roughly 60–65% of the variation in price, with typical predictions within about ₹9–10 lakh of the actual value.

Key Insights

- According to the Random Forest's feature importances, area is by far the strongest driver of price (close to half of total importance), followed at a distance by the number of bathrooms. Air conditioning, parking, and the number of stories form a second tier of moderately important features, while bedroom count and furnishing status matter comparatively little.
- Both models landed in a similar range: Linear Regression scored slightly better ($R^2 \approx 0.65$, MAE ≈ ₹9.7 lakh) than Random Forest ($R^2 \approx 0.61$, MAE ≈ ₹10.1 lakh). In plain terms, the model explains roughly 60–65% of why prices differ from one house to the next, and a typical prediction is off by about ₹9–10 lakh. That's useful for a rough estimate but not precise enough to replace a professional valuation.
- Bedroom count turned out to matter far less than expected once area was already in the model — area alone explains most of the price variation, and adding more bedrooms to a similarly-sized house barely moves the price. It was also notable that a simple Linear Regression held its own against Random Forest, suggesting the relationship between area/price is fairly linear for this dataset.
- Pricing and marketing should lead with total area and a short list of high-impact amenities (air conditioning, parking, number of stories/bathrooms) rather than bedroom count, since these are what the data shows buyers are actually paying a premium for. A model like this could give agents a quick first-pass price estimate before a full in-person appraisal.